



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to N.P.AGENCY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

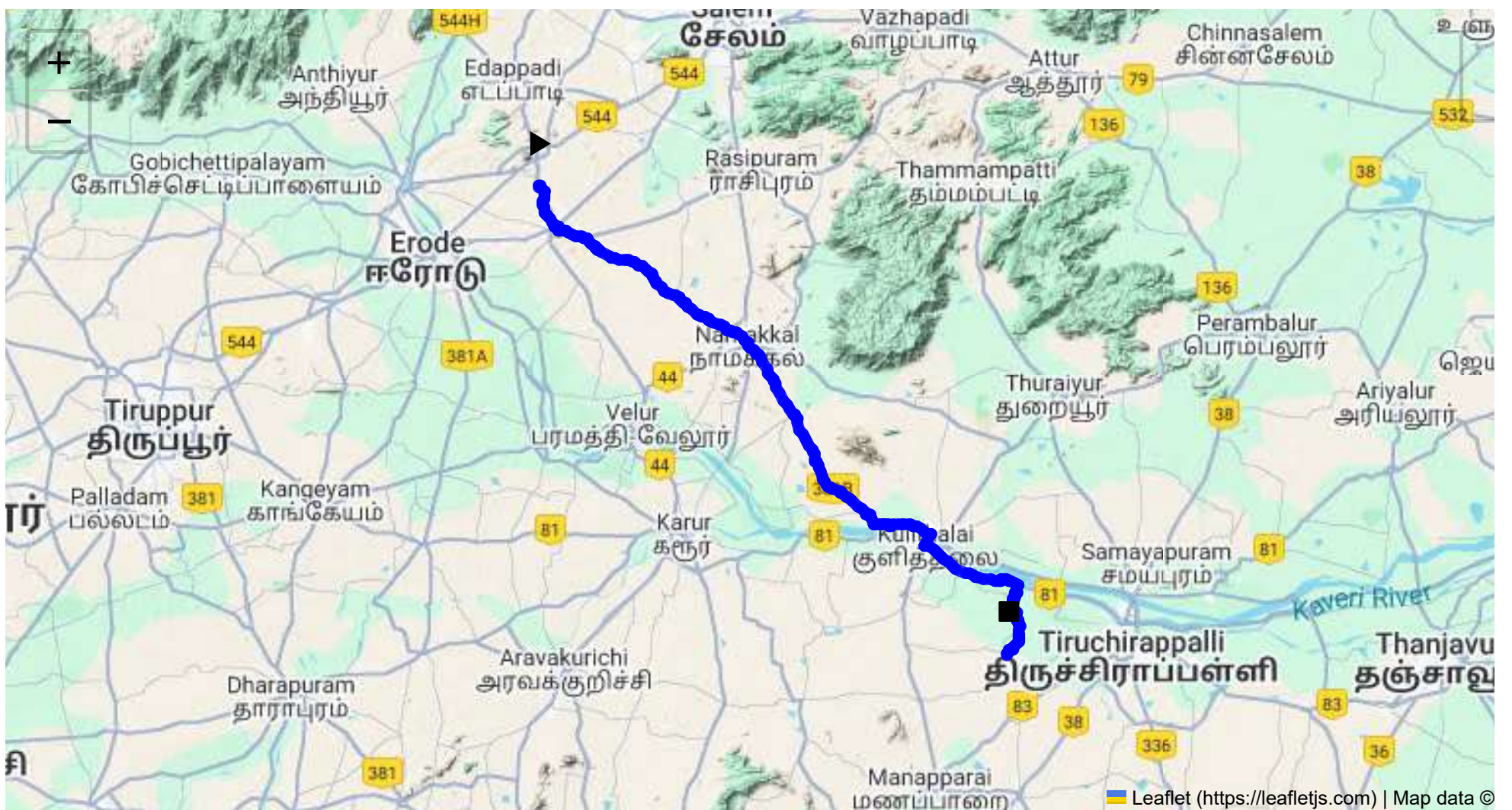
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 119.14 km
Estimated Duration: 2.5 hours
Adjusted Duration (Heavy Vehicle): 3.2 hours
Start: (11.4381, 77.8734)
End: (10.794012, 78.530451)



Welcome to the Journey Risk Management Study

Route Overview

The journey from CVQF+23W, Sangagiri, to QGVJ+J5 samathuvapuram spans approximately 119.14 kilometers and is typically managed in about 2.55 hours for heavy vehicles carrying hazardous materials. This route traverses through semi-urban and rural landscapes in Tamil Nadu, passing towns and villages that may have local traffic restrictions.

Weather Conditions and Hazards

The region experiences a tropical climate with hot, humid summers, and heavy rainfall during the monsoon season (June to September). Key weather-related hazards include:

- **Monsoon Rain:** Increased risk of road flooding and reduced visibility.
- **Heatwaves:** Potential for vehicle overheating during peak summer months.

Traffic Patterns

- **Peak Hours:** Morning (8 AM - 10 AM) and evening (5 PM - 7 PM) periods typically see heightened traffic, particularly near towns.
- **Congestion-Prone Areas:** Town centers and entry/exit points can become bottlenecks due to local traffic.

Road Quality and Infrastructure

- **Highway Sections:** Maintained with good surface quality, though occasional potholes may appear post-monsoon.
- **Rural Roads:** Narrower and potentially less well-maintained, requiring cautious navigation.
- **Infrastructure:** Basic to moderate, with some areas lacking advanced traffic management systems.

Alternative Routes

- **Emergency Detours:** A viable alternative is taking local state highways, though these may extend travel time. It's crucial to verify current conditions for any intended detour due to dynamic roadworks or closures.

Local Regulations

- **Hazardous Material Transport:** Ensure compliance with Tamil Nadu Motor Vehicles Act mandates. Restrictions often apply to transit through densely populated areas during peak times.

Historical Incidents

Past incident reviews indicate some issues with overturned vehicles due to sharp curves and narrow bridges. No specific trends of major hazardous material spills are noted, but caution is encouraged.

Environmental Considerations

The route passes agricultural zones and is close to some water bodies. Careful management of any material spills is crucial to prevent ecological impacts. Awareness of protected areas is necessary though there are no major wildlife sanctuaries nearby.

Communication Coverage

- **Network Strength:** Generally good along highways and townships, but potential dead zones exist in rural stretches. Consider using multiple carrier services for better reliability.

Emergency Response Times

- **Urban Areas:** Estimated response time is 20-30 minutes.
- **Rural Segments:** Potentially extended to 45-60 minutes due to distance from emergency services.

Summary of Risk Assessment

The overall risk of transporting hazardous materials along this route is moderate:

- **Main Risks:** Weather-induced challenges during monsoon, congestion delays in populated areas.
- **Recommendations:** Enhanced navigation systems to monitor live traffic and weather conditions, regular maintenance checks especially before entering/prior to long distances to avoid mechanical failures, and consultations with local safety authorities for updated regulations or advisories.

Overall, the preparation and adjustment to real-time conditions are crucial for ensuring safe passage, given the varied terrain and environmental conditions along this route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44019, 77.87547	15 KM/Hr
3	Turn	Medium	11.39685, 77.88861	30 KM/Hr
4	Turn	Medium	11.38376, 77.89914	30 KM/Hr
5	Turn	High	11.38186, 77.90258	15 KM/Hr
6	Blind Spot	Blind Spot	11.37841, 77.90140	10 KM/Hr
7	Turn	Medium	11.33425, 78.00843	30 KM/Hr
8	Turn	Medium	11.22411, 78.16637	30 KM/Hr
9	Turn	Medium	11.21992, 78.16759	30 KM/Hr
10	Turn	Medium	11.12316, 78.23249	30 KM/Hr
11	Turn	Medium	11.12001, 78.23557	30 KM/Hr
12	Turn	Medium	10.98686, 78.33767	30 KM/Hr
13	Turn	Medium	10.97256, 78.34408	30 KM/Hr
14	Turn	High	10.96031, 78.42222	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
15	Turn	High	10.96084, 78.42307	15 KM/Hr
16	Blind Spot	Blind Spot	10.96050, 78.42371	10 KM/Hr
17	Blind Spot	Blind Spot	10.94455, 78.41419	10 KM/Hr
18	Turn	High	10.94435, 78.41548	15 KM/Hr
19	Turn	Medium	10.94465, 78.41565	30 KM/Hr
20	Turn	Medium	10.94478, 78.41595	30 KM/Hr
21	Turn	Medium	10.89656, 78.52077	30 KM/Hr
22	Turn	Medium	10.89694, 78.52119	30 KM/Hr
23	Blind Spot	Blind Spot	10.88965, 78.54586	10 KM/Hr
24	Turn	High	10.88043, 78.54458	15 KM/Hr
25	Turn	High	10.88035, 78.54354	15 KM/Hr
26	Turn	High	10.87949, 78.54326	15 KM/Hr
27	Turn	High	10.87931, 78.54275	15 KM/Hr
28	Turn	Medium	10.87339, 78.54096	30 KM/Hr
29	Turn	Medium	10.87090, 78.53973	30 KM/Hr
30	Turn	High	10.87065, 78.53930	15 KM/Hr
31	Turn	Medium	10.86866, 78.53904	30 KM/Hr
32	Turn	High	10.86817, 78.53870	15 KM/Hr
33	Turn	High	10.86814, 78.53816	15 KM/Hr
34	Turn	High	10.86518, 78.53852	15 KM/Hr
35	Turn	Medium	10.86079, 78.53641	30 KM/Hr
36	Turn	Medium	10.85759, 78.53682	30 KM/Hr
37	Turn	Medium	10.85289, 78.54270	30 KM/Hr
38	Turn	Medium	10.83579, 78.54689	30 KM/Hr
39	Turn	High	10.83526, 78.54797	15 KM/Hr
40	Turn	Medium	10.81952, 78.54621	30 KM/Hr
41	Turn	Medium	10.81876, 78.54657	30 KM/Hr
42	Turn	Medium	10.81825, 78.54727	30 KM/Hr
43	Turn	Medium	10.81722, 78.54749	30 KM/Hr
44	Turn	Medium	10.81224, 78.54628	30 KM/Hr
45	Turn	Medium	10.80332, 78.53768	30 KM/Hr
46	Turn	High	10.80016, 78.53726	15 KM/Hr
47	Turn	High	10.79972, 78.53334	15 KM/Hr
48	Turn	Medium	10.79864, 78.53300	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
49	Turn	Medium	10.79497, 78.53044	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	Jayaa Hospital	11.3848223, 77.9021886	30 km/h	Medium
6	hospital	Vivekananda Medical Care Hospital	11.364562, 77.9482668	30 km/h	Medium
10	hospital	Manickampalayam, Government Hospital	11.32887, 78.017713	30 km/h	Medium
15	hospital	Aruthra Hospital	11.2264666, 78.1631524	30 km/h	Medium
16	clinic	Sri Ram Nursing Home, Namakkal	11.2259986, 78.1631944	30 km/h	Medium
17	clinic	Thangavel Skin Care	11.2253765, 78.1644957	30 km/h	Medium
18	hospital	Rudhra Hospital	11.226569, 78.163216	30 km/h	Medium
19	hospital	Rajendran Children's Hospital	11.2251887, 78.1649142	30 km/h	Medium
20	clinic	Lung Care Centre	11.2276789, 78.1638586	30 km/h	Medium
21	hospital	Vignesh Hospital	11.224818, 78.1621315	30 km/h	Medium
22	hospital	Venkat Hospital	11.2253645, 78.1646159	30 km/h	Medium
23	hospital	Shree Meenakshi Dental Hospital	11.225269, 78.164275	30 km/h	Medium
24	hospital	Ilamurugan Hospital	11.2248399, 78.166586	30 km/h	Medium
25	hospital	Annai Eye Hospital	11.2241645, 78.1641366	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
29	clinic	Kumaran Poly Clinic	11.2233138, 78.1639212	30 km/h	Medium
30	clinic	Chandra Test Tube Baby Centre	11.2241806, 78.1645931	30 km/h	Medium
31	hospital	Aburvaa Ayurveda Hospital	11.2228807, 78.1640453	30 km/h	Medium
32	clinic	Vishnu Ayurveda Clinic	11.224262, 78.164173	30 km/h	Medium
33	clinic	Dhanvanthiri Physiotheraphy	11.2245256, 78.1638528	30 km/h	Medium
34	hospital	Aravinth Hospital	11.2233336, 78.1672805	30 km/h	Medium
35	hospital	Sri Nidhi Hospital, Namakkal	11.2255711, 78.1687291	30 km/h	Medium
36	clinic	Athithya Nursing Home	11.2237726, 78.1686447	30 km/h	Medium
37	hospital	Kathiravan Hospital	11.226651, 78.167351	30 km/h	Medium
38	hospital	Sri Ramana Hospital	11.223503, 78.1685433	30 km/h	Medium
39	hospital	Vanitha Child Care & Skin Hospital	11.2243378, 78.16811	30 km/h	Medium
40	hospital	Sri Dhanvanthries Hospital	11.2241181, 78.1678614	30 km/h	Medium
41	hospital	Nirmala Hospital	11.2179361, 78.1665001	30 km/h	Medium
42	hospital	Surya Hospital, Namakkal	11.2178506, 78.1660375	30 km/h	Medium
43	hospital	Shanthi Hospitals, Namakkal	11.2190661, 78.1685735	30 km/h	Medium
44	hospital	Ganga Hospital	11.21742, 78.170272	30 km/h	Medium
45	hospital	Government Hospital	11.2168924, 78.169068	30 km/h	Medium
47	police	Police Station	11.2185353, 78.1695347	30 km/h	Medium
48	police	Women Police Station	11.2183133, 78.1692677	30 km/h	Medium
49	hospital	Manoharan Hospital	11.218016, 78.171629	30 km/h	Medium
50	hospital	Saravana Hospital	11.216652, 78.171298	30 km/h	Medium
52	hospital	Latha Eye Hospital	11.2185689, 78.1719035	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
53	hospital	Vasan Eye Care Hospital, Namakkal	11.216301, 78.1717694	30 km/h	Medium
54	hospital	Sri Ragavendra Hospital	11.2151734, 78.1703431	30 km/h	Medium
55	hospital	Aishwarya Hospital	11.2164042, 78.1715972	30 km/h	Medium
56	hospital	Thangam Hospital,Namakkal	11.21472, 78.171857	30 km/h	Medium
57	hospital	Poornima Hospital	11.215533, 78.16911	30 km/h	Medium
58	hospital	Nagarajan Hospital	11.2171673, 78.1718618	30 km/h	Medium
59	clinic	Ramesh ENT & Skin Care	11.217133, 78.171071	30 km/h	Medium
60	hospital	SMK Hospital	11.2172759, 78.1706866	30 km/h	Medium
61	clinic	ENT Clinic	11.215786, 78.170545	30 km/h	Medium
62	clinic	Hari Chest Clinic	11.2162053, 78.1710093	30 km/h	Medium
63	hospital	Dr. Kandasamy Hospital	11.216051, 78.172052	30 km/h	Medium
64	clinic	Primecare Diabetic Clinic	11.213992, 78.1737091	30 km/h	Medium
65	clinic	Dr. Shanmugam ENT Clinic	11.218762, 78.171206	30 km/h	Medium
66	clinic	Children's Clinic	11.217004, 78.171504	30 km/h	Medium
67	clinic	Krishiv's Dental Care & Orthodontic Centre	11.2184886, 78.1713346	30 km/h	Medium
71	hospital	Dr. A. Nagarajan Children's Hospital	11.212765, 78.1747068	30 km/h	Medium
72	hospital	M.M. Hospital	11.2103089, 78.1776545	30 km/h	Medium
73	clinic	Ram Clinic	11.2090922, 78.1790782	30 km/h	Medium
74	clinic	Prasith Clinic	11.208902, 78.180957	30 km/h	Medium
75	hospital	Imayam Speciality Hospital	11.2078899, 78.1797861	30 km/h	Medium
76	clinic	Aburvaa Multi Special Herbal Clinic	11.207568, 78.183483	30 km/h	Medium
77	hospital	Sugam Children's Hospital	11.2060083, 78.1854033	30 km/h	Medium
79	hospital	Government Hospital, Valayapatty	11.1212591, 78.2315962	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
82	hospital	Saravana Hospitals	10.9944313, 78.3273838	30 km/h	Medium
83	hospital	Dr. Siva Hospital	10.9877652, 78.3359228	30 km/h	Medium
84	hospital	Dr. Alva Hospital	10.987225, 78.336154	30 km/h	Medium
85	hospital	Ravikumar Hospital	10.988317, 78.336248	30 km/h	Medium
86	police	Thottiyam Police station	10.9874415, 78.3363001	30 km/h	Medium
87	hospital	Government Hospital, Thottiyam	10.98388, 78.338648	30 km/h	Medium
88	hospital	Government Hospital	10.9428055, 78.4146362	30 km/h	Medium
89	hospital	T M R Hospital	10.9429825, 78.4158242	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
7	college	Vivekanandha Educational Institutions for Women	11.3596633, 77.9435556	30 km/h	Medium
8	college	Vivekanandha Dental College	11.3643705, 77.9475897	30 km/h	Medium
9	college	Vivekanandha College of engineering for Women	11.3565938, 77.9465342	30 km/h	Medium
11	college	CMS College Of Engineering	11.2647508, 78.0969044	30 km/h	Medium
12	college	The Surabi College of Polytechnic	11.2660726, 78.0939552	30 km/h	Medium
13	school	Porasapalayam Government School	11.2504918, 78.1163024	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
14	school	Sri Vinayaga Higher Secondary School	11.2479837, 78.1182908	30 km/h	Medium
26	school	Government Girls School	11.2238954, 78.1662583	30 km/h	Medium
27	school	Government Elementary School	11.224555, 78.163967	30 km/h	Medium
28	school	Government Secondary School	11.2244426, 78.1647533	30 km/h	Medium
46	school	Government Girls Higher Secondary School	11.2179186, 78.1672383	30 km/h	Medium
51	college	Veterinary College and Research Institute Hospital	11.2144944, 78.1722653	30 km/h	Medium
68	college	Namakkal Kavignar Ramalingam Government Arts College for Women	11.2125137, 78.1724654	30 km/h	Medium
69	school	Kandasamy Kandar Matriculation and Higher Secondary School	11.2137988, 78.1698234	30 km/h	Medium
70	school	Selvam Matriculation Higher Secondary School	11.2124286, 78.1770824	30 km/h	Medium
78	college	King college	11.1692622, 78.2037212	30 km/h	Medium
80	school	Government Higher Secondary School	11.0234395, 78.2859835	30 km/h	Medium
81	college	Vetri vinayaga institution	11.013503, 78.3072053	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr
Coordinates: 11.38395, 77.89480



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44019, 77.87547



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.39685, 77.88861



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.38376, 77.89914



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38186, 77.90258



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.37841, 77.90140



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.33425, 78.00843



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.22411, 78.16637



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.21992, 78.16759



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.12316, 78.23249



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.12001, 78.23557



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.98686, 78.33767



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.97256, 78.34408



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.96031, 78.42222



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.96084, 78.42307



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 10.96050, 78.42371



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 10.94455, 78.41419



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.94435, 78.41548



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.94465, 78.41565



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.94478, 78.41595



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.89656, 78.52077



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.89694, 78.52119



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 10.88965, 78.54586



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.88043, 78.54458



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.88035, 78.54354



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.87949, 78.54326



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.87931, 78.54275



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.87339, 78.54096



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.87090, 78.53973



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.87065, 78.53930



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.86866, 78.53904



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.86817, 78.53870



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.86814, 78.53816



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.86518, 78.53852



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.86079, 78.53641



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.85759, 78.53682



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.85289, 78.54270



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.83579, 78.54689



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.83526, 78.54797
